System Design Specification (SDS)

Modular Chilled-Water Cooling Plant with Building Management System Integration

ISA/IEC 62443 Target Security Level 3 (SL-T 3) — design of FRS-CHW-001

Template with worked example

Worked example: data center plant — Trane Symbio 800 chiller modules · Rockwell ControlLogix 5580 plant control · Siemens Desigo CC (primary) · Kepware historian (secondary, long-term)

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Vendor settings, product capabilities and catalog references are as of 28 October 2026; confirm them at FAT, SAT and handover.

Using this template

- Each design element names the FRS rows it implements; Annex A proves every FRS row is designed in this SDS or in the HDS.
- Design decisions record the options considered and why one was chosen, so a project can change a decision knowingly.
- Addresses, VLANs and ports are worked-example values; replace them with the project plan.
- Black text is the template requirement. *Blue italic text is the worked example.* **[Bracketed blue text]** is a project fill-in with the worked-example default.
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Approvals

Role	Name	Signature	Date
Asset owner — OT cybersecurity	**[Name]**		**[Date]**
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Revision history

Rev	Date	Description	Author
[A]	**[Date]**	**[Initial issue for review]**	**[Name]**

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1 Purpose and design principles

This System Design Specification (SDS) defines how the functions in FRS-CHW-001 are designed: architecture, network and conduit design, control software structure, BMS and historian design, security services and per-product security configuration baselines. The Hardware Design Specification (HDS) covers the physical hardware; the V&V, commissioning and acceptance work instruction tests both.

Design principles: control stays local and autonomous (chiller, plant controller, skid); supervision is read-mostly; secure protocol variants are used wherever both ends support them; every insecure protocol is confined; SL-A 3 comes from the configured system, with product certification as supporting evidence only.

2 Design decisions

ID	Decision	Options considered	Chosen design and rationale	FRS
DD-01	**Plant controller architecture**	(1) One redundant ControlLogix pair (1756-RM2). (2) Two independent controllers A and B, each owning half of the equipment, with plant-master role failover in logic.	*Option 2. Each controller keeps its native OPC UA server for the primary BMS path; OPC UA support under controller redundancy is unconfirmed (OI-03). Losing one controller loses only the units it owns, and the surviving controller takes the plant-master role (staging, TES, economizer) within 10 s. Option 1 remains acceptable if Rockwell confirms OPC UA with redundancy.*	FRS-DEG-003, FRS-CHL-006, FRS-INT-003, FRS-CMP-004
DD-02	**Plant-to-BMS integration**	(1) Controller-native OPC UA. (2) Data server only. (3) BACnet gateway.	*Controller-native OPC UA to Desigo CC (primary); Kepware OPC UA client as read-only secondary feeding the Kepware historian (owner decision).*	FRS-INT-003, FRS-INT-004, FRS-HMI-007
DD-03	**Chiller integration**	(1) Modbus TCP via MVI56E-MNETC. (2) BACnet/IP to controller via gateway.	*Option 1. Each controller polls its own two chillers with an MVI56E-MNETC; writes only from the owning controller; Desigo CC reads chillers over BACnet/SC when Trane supports it, BACnet/IP read-only until then.*	FRS-INT-001, FRS-INT-002, FRS-CHL-008
DD-04	**Zone boundaries**	(1) One firewall pair for all conduits. (2) Plant boundary pair plus IDMZ pair.	*Option 2. FortiGate Rugged 70F HA pair at the plant boundary (Z1–Z4, Z10 ↔ Z5, Z6, Z7); FortiGate 200F HA pair forming the IDMZ (Z8) between OT and Z9.*	FRS-NET-001, FRS-NET-002, FRS-NET-004, FRS-NET-005, FRS-SEC-502
DD-05	**Plant control network**	(1) Star. (2) DLR ring. (3) PRP.	*DLR ring of Stratix 5800 switches in Z2 with each controller, chiller module switch and drive group on its own ring node; recovery inside the 100 ms limit.*	FRS-NET-007, FRS-DEG-004
DD-06	**Virtualization**	(1) One cluster, port-group separation. (2) Separate host pairs per zone.	*Option 2. Two ESXi hosts for Z5 (Desigo CC, Kepware), two for Z7 (infrastructure services); hypervisor management only in Z7.*	FRS-SEC-504, FRS-NET-008
DD-07	**Identity**	(1) Enterprise domain. (2) Separate OT domain.	*Separate OT Active Directory forest in Z7 with no trust to enterprise; Desigo CC, FactoryTalk, Windows hosts, FortiGate and Stratix authenticate to it.*	FRS-SEC-101, FRS-SEC-103, FRS-SEC-104, FRS-SEC-107
DD-08	**PKI**	(1) Vendor self-signed. (2) Owner two-tier PKI.	*Offline root CA (powered off, stored in a safe) and online issuing CA in Z7; CRL published inside OT.*	FRS-SEC-108, FRS-SEC-109, FRS-CMP-108, FRS-CMP-109
DD-09	**Time**	(1) Enterprise NTP. (2) Local GPS sources.	*Two GPS-disciplined NTP appliances in Z7; Stratix 5800 PTP boundary clock in Z2 for CIP Sync to the controllers.*	FRS-SEC-211, FRS-CMP-211
DD-10	**Local operator interface**	(1) Panel-mounted HMI. (2) FactoryTalk View SE station in Z2.	*FactoryTalk View SE local station on a Cincoze CV-100 + P2102 panel PC in Z2, connected to both controllers; independent of Desigo CC.*	FRS-HMI-001, FRS-DEG-001
DD-11	**Remote access**	(1) Vendor cloud. (2) Owner jump host.	*Owner jump host in Z8 with MFA, per-session approval and recording; all vendor cloud connectors disabled.*	FRS-SEC-113, FRS-INT-011
DD-12	**Historian**	(1) Desigo CC only. (2) Kepware historian long-term.	*Kepware historian in Z5 on an encrypted volume as the long-term store (7 years); outbound-only forwarding to campus aggregation through Z8.*	FRS-HMI-006, FRS-HMI-007, FRS-DEG-002
DD-13	**Plant-room HVAC**	(1) Plant controller. (2) Desigo PXC.	*Desigo PXC7 on BACnet/SC in its own zone Z10; refrigerant emergency ventilation stays hardwired.*	FRS-PRH-001, FRS-INT-008, FRS-SAF-001

3 Network architecture and addressing

Zones and conduits are as defined in the URS (section 6) and FRS (section 7).

Zone	Name	VLAN	Subnet	Boundary
Z1-1 … Z1-4	Chiller modules 1–4	[VLAN 211–214]	[10.20.11.0/28 … 10.20.14.0/28]	Plant boundary FortiGate

Zone	Name	VLAN	Subnet	Boundary
Z2	Plant control	[VLAN 220]	[10.20.20.0/24]	Plant boundary FortiGate
Z3	Plant electrical metering	[VLAN 230]	[10.20.30.0/26]	Plant boundary FortiGate
Z4	Vendor packages	[VLAN 240]	[10.20.40.0/26]	Plant boundary FortiGate
Z10	Plant-room HVAC	[VLAN 300]	[10.20.100.0/27]	Plant boundary FortiGate
Z5	BMS supervisory	[VLAN 250]	[10.20.50.0/26]	Plant boundary and IDMZ FortiGates
Z6	Engineering	[VLAN 260]	[10.20.60.0/27]	Plant boundary and IDMZ FortiGates
Z7	OT infrastructure (services)	[VLAN 270]	[10.20.70.0/26]	Plant boundary and IDMZ FortiGates
Z7-M	OT infrastructure (management)	[VLAN 271]	[10.20.71.0/26]	Plant boundary FortiGate
Z8	Industrial DMZ	[VLAN 280]	[10.20.80.0/27]	IDMZ FortiGate pair

The Z2 plant control network is a DLR ring of Stratix 5800 switches (DD-05); chiller module switches and drive groups attach to ring nodes. The Z5, Z6 and Z7 networks run on Catalyst 9300 switches. Every inter-zone path crosses a FortiGate interface; no host is dual-homed across OT zones (FRS-NET-001).

4 Conduit rule register (worked example)

Every rule is deny-by-default plus explicit allow; each allow names its FRS justification. Protocol-aware restrictions (read-only function codes and services) apply where the FortiGate signature set supports them (OI-14).

Rule	Conduit	Source	Destination	Service	Action	Justification
R-C1-01	C1-1, C1-2	*Controller A MVI56E-MNETC*	*Symbio 800 #1, #2*	*TCP 502*	Allow	FRS-INT-001
R-C1-02	C1-3, C1-4	*Controller B MVI56E-MNETC*	*Symbio 800 #3, #4*	*TCP 502*	Allow	FRS-INT-001
R-C9-01	C9-n	*Desigo CC*	*Symbio 800 #1–#4*	*BACnet/SC TCP 443 (when available); else UDP 47808 read-only*	Allow	FRS-INT-002
R-C2-01	C2	*Desigo CC servers*	*Controllers A, B*	*TCP 4840 (OPC UA)*	Allow	FRS-INT-003
R-C2-02	C2	*Kepware*	*Controllers A, B*	*TCP 4840 (OPC UA)*	Allow	FRS-INT-004
R-C3-01	C3	*EWS*	*Controllers A, B; PowerFlex 755T*	*TCP 2221 / UDP 2221 (CIP Security)*	Allow	FRS-INT-010, FRS-LCM-004
R-C3-02	C3	*EWS*	*Symbio 800, PowerLogic, QuickServer, S7-1500, Desigo PXC*	*HTTPS 443; ION TLS 7443; vendor tool ports per OI-20*	Allow	FRS-LCM-004
R-C4-01	C4	*Desigo CC, Kepware*	*PM8000*	*UDP 47808 (read services only where filterable)*	Allow	FRS-INT-009
R-C4-02	C4	*Desigo CC, Kepware*	*PM5560*	*TCP 502 (read function codes only where filterable)*	Allow	FRS-INT-009
R-C5-01	C5	*Controllers A, B MVI56E-MNETC*	*S7-1500 skid*	*TCP 502 (read)*	Allow	FRS-INT-006
R-C10-01	C10	*Desigo CC, Kepware*	*S7-1500 skid*	*TCP 4840 (OPC UA)*	Allow	FRS-INT-005
R-C10-02	C10	*Desigo CC, Kepware*	*QuickServer*	*UDP 47808 (read)*	Allow	FRS-INT-007
R-C11-01	C11	*Desigo PXC*	*Desigo CC BACnet/SC hub*	*TCP 443 (BACnet/SC)*	Allow	FRS-INT-008
R-C6-01	C6	*Jump host (Z8)*	*EWS, Desigo CC clients*	*RDP 3389 (approved sessions only)*	Allow	FRS-SEC-113

Rule	Conduit	Source	Destination	Service	Action	Justification
R-C6-02	C6	*Patch / AV staging (Z8)*	*Z5, Z6, Z7 update services*	*HTTPS 443 (pull from Z8 by owner services)*	Allow	FRS-SEC-302
R-C12-01	C12	*Kepware historian (Z5)*	*Campus aggregation forwarder (Z8)*	*Per SDS 7 (OPC UA or HTTPS), outbound only*	Allow	FRS-INT-011
R-C7-01	C7	*Owner and vendor users (Z9)*	*Jump host*	*HTTPS 443 with MFA*	Allow	FRS-SEC-113
R-C8-01	C8	*All OT zones*	*NTP servers (Z7)*	*UDP 123*	Allow	FRS-SEC-211
R-C8-02	C8	*All OT zones*	*Log collector (Z7)*	*TCP 6514 (syslog over TLS); UDP 514 only for devices without TLS (listed)*	Allow	FRS-SEC-208
R-C8-03	C8	*Z5, Z6, Z7 hosts; FortiGate; Stratix*	*Domain controllers (Z7)*	*LDAPS 636, Kerberos 88, RADIUS 1812, TACACS+ 49*	Allow	FRS-SEC-103
R-C8-04	C8	*Management hosts (Z7-M)*	*Switches, FortiGates, ESXi*	*SSH 22, HTTPS 443, SNMPv3 UDP 161*	Allow	FRS-NET-008
R-C8-05	C8	*Backup server (Z7)*	*Z5, Z6 hosts; FortiGate; Stratix*	*Per backup product; SCP or SFTP for devices*	Allow	FRS-SEC-703
R-DEF	All	*Any*	*Any*	*Any*	Deny and log	FRS-NET-002

5 Control software design

Plant controller program for ControlLogix 5580 controllers A and B, written to the Secure PLC Coding Top 20 (FRS section 10).

ID	Element	Design (worked example)	FRS
SW-01	**Task structure**	*Periodic tasks: Fast 50 ms (I/O conditioning, interlocks, paired-I/O checks); Sequence 250 ms (equipment state machines); Plant 1 s (staging, TES, economizer, master role, calculations); Comms 500 ms (MVI56E-MNETC and OPC UA blocks). Power-up handler and controller fault handler programmed.*	FRS-CHL-*, FRS-HYD-*, FRS-CT-*, FRS-ECO-*, FRS-TES-*, FRS-PLC-015, FRS-PLC-016, FRS-PLC-018
SW-02	**Equipment add-on instructions**	*AOIs: CHL (chiller via Modbus), PMP (pump via CIP), FAN (tower fan), VLV (modulating and two-position valve), FLW (flow and thermal calc), VAL (write validation), ALM (alarm with priority, deadband, delay, shelving). Each AOI version-controlled with a test record.*	FRS-PLC-001, FRS-PLC-006, FRS-PLC-007, FRS-PLC-011, FRS-PLC-012, FRS-PLC-020, FRS-MET-002
SW-03	**Plant-master role**	*Controllers A and B exchange produced/consumed heartbeat tags; A holds the plant-master role by default. If A's heartbeat is lost for 5 s, B takes the role within 10 s using the last shared plant state; role changes are alarmed P2 and logged.*	FRS-DEG-003, FRS-CHL-006, FRS-PLC-004
SW-04	**BMS interface UDTs**	*BMS_RO (read-only status, alarms), BMS_WRQ (write requests: plant mode, CHW setpoint, demand limit, alarm acknowledge), BMS_VAL (validated values used by logic). Only BMS_WRQ has External Access Read/Write and is exposed for writing on OPC UA; all other tags None or Read Only.*	FRS-PLC-008, FRS-PLC-010, FRS-PLC-014, FRS-INT-003, FRS-SEC-305
SW-05	**Write-request handling**	*Each request carries value and request counter; VAL AOI checks range, rate, mode and role flag, copies accepted values to BMS_VAL, returns status codes, and logs rejected requests with the OPC UA user name supplied by the server diagnostics where available.*	FRS-CHL-001, FRS-CHL-007, FRS-SEC-212, FRS-PLC-008
SW-06	**Modes and fail states**	*Mode state machine per FRS-MOD; retentive mode and setpoint tags; output hold until first full scan validates inputs; drive and valve fail states per FRS-DEG table.*	FRS-MOD-*, FRS-DEG-*, FRS-PLC-015, FRS-SEC-306
SW-07	**Integrity and diagnostics**	*Change detection mask set to logic, tag-value, firmware and mode changes; Audit Value compared with the released value each scan; keyswitch position, controller mode, uptime, scan time, memory use and major faults sent to BMS and trended.*	FRS-PLC-002, FRS-PLC-005, FRS-PLC-016, FRS-PLC-017, FRS-PLC-018, FRS-PLC-019, FRS-SEC-304, FRS-HMI-005

ID	Element	Design (worked example)	FRS
SW-08	**Program protection and release**	*Source protection on released AOIs; firmware and project revision recorded; .ACD and .L5X archived with SHA-256 at each release; MOC number in every change comment.*	FRS-PLC-005, FRS-LCM-003, FRS-SEC-703
SW-09	**Communication hygiene**	*Unused controller services off; MVI56E-MNETC client map limited to listed registers; no Modbus server enabled; OPC UA anonymous off; CIP Security policy from FactoryTalk Policy Manager.*	FRS-PLC-013, FRS-PLC-014, FRS-PLC-009, FRS-PLC-003, FRS-SEC-707
SW-10	**Skid program (vendor)**	*Vendor applies the Top 20 to the S7-1500 program, read-only Modbus data area for the plant, OPC UA server per FRS-INT-005, and declares compliance.*	FRS-PLC-021, FRS-WT-*
SW-11	**Metering and plant calculations**	*Thermal load in FLW AOI; efficiency calculated in Desigo CC; meter polling per FRS-MET-001.*	FRS-MET-*
SW-12	**Safety separation**	*No safety function is implemented in software. Plant controller monitors refrigerant, fire and permissive signals only; no program path writes to a safety device.*	FRS-SAF-*
SW-13	**Lifecycle records**	*Inventory, spares baseline and MOC records maintained in Z7 per SDS 8.*	FRS-LCM-*

6 BMS design (Desigo CC)

ID	Element	Design (worked example)	FRS
BMS-01	**Server topology**	*Desigo CC server VM (Windows Server 2026) on the Z5 ESXi pair with a standby option per Siemens supported configuration (confirm); clients on Windows 11 operator stations in Z5.*	FRS-HMI-002, FRS-SEC-504
BMS-02	**Users and rights**	*Desigo CC user groups mapped to OT domain groups per the role matrix (SDS 10); scope rights restrict commands to the BMS_WRQ allowlist.*	FRS-SEC-201, FRS-SEC-101, FRS-HMI-002
BMS-03	**Integration drivers**	*OPC UA client connections to controllers A and B and to the S7-1500 (SignAndEncrypt, owner-PKI certificates); BACnet/SC hub for Desigo PXC and, when available, Symbio 800; BACnet/IP read-only for PM8000, QuickServer and Symbio fallback; Modbus TCP read-only for PM5560 (or via Kepware).*	FRS-INT-002, FRS-INT-003, FRS-INT-005, FRS-INT-007, FRS-INT-008, FRS-INT-009, FRS-SEC-301
BMS-04	**Alarms**	*Four priorities per FRS-HMI-003; alarm attributes from the controller ALM AOI; shelving restricted to Shift Supervisor; security-relevant alarms per FRS-HMI-005.*	FRS-HMI-003, FRS-HMI-004, FRS-HMI-005
BMS-05	**Trends and reports**	*1-min trends for key analogs, change-of-value for digitals, 13-month retention; daily performance, monthly energy and weekly security reports.*	FRS-HMI-006, FRS-HMI-008, FRS-MET-003
BMS-06	**Graphics**	*Plant overview, chiller, hydronic, tower, economizer, TES, skid, plant-room HVAC and network-health pages; command buttons only for allowlisted requests.*	FRS-HMI-002

7 Historian design (Kepware)

ID	Element	Design (worked example)	FRS
HIS-01	**Kepware project**	*Channels: OPC UA client to controllers A and B and to the S7-1500; BACnet/IP client to PM8000 and QuickServer; Modbus TCP client to PM5560. All tags read-only (client write disabled per channel where supported, and read-only OPC UA user).*	FRS-INT-004, FRS-INT-005, FRS-INT-009, FRS-HMI-007
HIS-02	**Kepware historian storage**	*Critical tags at 1 s, others at 10 s; 7-year retention; storage on an encrypted volume; sizing per OI-17.*	FRS-HMI-007, FRS-SEC-401
HIS-03	**Campus aggregation**	*Outbound-only forwarding from Kepware to the campus aggregation forwarder in Z8 over OPC UA or HTTPS; local buffering during link loss.*	FRS-INT-011, FRS-DEG-002

ID	Element	Design (worked example)	FRS
HIS-04	**Kepware hardening**	*Runtime as a least-privilege service account; user manager with named admins; OPC UA server endpoints (if enabled) SignAndEncrypt only; unused drivers and interfaces removed.*	FRS-SEC-201, FRS-SEC-707, FRS-DEV-S204, FRS-DEV-S302

8 Security services design

ID	Element	Design (worked example)	FRS
SVC-01	**Identity**	*OT AD forest (two DCs, Z7); groups per role; GPOs for password, lockout, banner, screen lock, USB control, application control, audit policy; Windows LAPS; FortiGate and Stratix via RADIUS and TACACS+; jump host MFA.*	FRS-SEC-1*, FRS-SEC-201, FRS-SEC-203, FRS-SEC-204, FRS-SEC-205, FRS-SEC-206, FRS-SEC-207
SVC-02	**PKI**	*Two-tier PKI per DD-08; certificate templates for TLS server, OPC UA application, BACnet/SC device and CIP Security device; validity 2 years; renewal alarm at 60 days.*	FRS-SEC-108, FRS-SEC-109, FRS-SEC-403, FRS-CMP-1*
SVC-03	**Time**	*Per DD-09; NTP authentication keys on Stratix and FortiGate; offset monitoring.*	FRS-SEC-211
SVC-04	**Logging**	*Central collector in Z7 receiving syslog over TLS where supported, Windows event forwarding and ESXi syslog; source-silent detection; 12 months online, 3 years archive; read-only API for the owner monitoring function.*	FRS-SEC-208, FRS-SEC-209, FRS-SEC-210, FRS-SEC-309, FRS-SEC-601, FRS-CMP-2*, FRS-CMP-3*, FRS-CMP-6*
SVC-05	**Malicious code protection**	*Vendor-qualified endpoint protection or application control on Desigo CC, Kepware, EWS and operator hosts; media scanning station in Z7; FortiGate AV and IPS at C6 and C7.*	FRS-SEC-302, FRS-DEV-H302
SVC-06	**Backup and recovery**	*Backup server in Z7; schedule per FRS-SEC-703; quarterly restore on the test bench; gold images and recovery runbooks per module.*	FRS-SEC-703, FRS-SEC-704, FRS-SEC-705, FRS-CMP-7*
SVC-07	**Monitoring**	*Passive OT network monitoring sensors on SPAN ports of the Z1–Z5 and Z10 switches; console in Z7; alerts to the owner security operations function.*	FRS-SEC-602, FRS-SEC-202
SVC-08	**Patch and change management**	*Vendor-qualified patches staged in Z8; applied module by module under MOC; baseline comparison after change.*	FRS-SEC-303, FRS-SEC-706, FRS-SEC-708, FRS-DEV-H310, FRS-DEV-E310, FRS-DEV-N310
SVC-09	**Data protection**	*BitLocker on hosts; encrypted backups; decommissioning procedure; cipher policy.*	FRS-SEC-4*, FRS-CMP-4*
SVC-10	**Segmentation and boundaries**	*Per SDS 3 and 4; island-mode policy sets per zone; FortiGate HA without bypass; application control blocks person-to-person protocols; DoS policies per conduit.*	FRS-SEC-5*, FRS-SEC-701, FRS-SEC-702, FRS-NET-*, FRS-CMP-5*, FRS-DEV-N*
SVC-11	**Physical and component hardening**	*Per-product baselines in SDS 9; physical controls detailed in the HDS.*	FRS-DEV-E*, FRS-DEV-H*, FRS-CMP-0*, FRS-SEC-307, FRS-SEC-308

9 Security configuration baselines per product

Highlights only; the full baseline for each product is its per-product ISA/IEC 62443 SL-3 checklist, which is also the V&V procedure basis. Exported configurations are compared with these baselines at SAT and after every change (FRS-SEC-706).

Product	Baseline highlights (worked example)	Checklist
Trane Symbio 800	*Named Symbio UI users with permissions; HTTPS-only web access; web UI over Ethernet limited to Z6 addresses; Trane Connect off; Air-Fi, Wi-Fi and cellular not fitted; unused protocols (LON) off; BAS communication-loss behaviour set (OI-12); automated encrypted backup on; NTP set.*	Symbio 800 (EDR)
ControlLogix 5580 (A, B)	*Keyswitch RUN with key held by Controls Engineer; FactoryTalk Security; CIP Security via 1756-EN4TR; change detection mask and Audit Value monitoring; web pages off; front USB and SD disabled where supported (OI-13); OPC UA server SignAndEncrypt, anonymous off; External Access per SW-04; CIP Sync time.*	ControlLogix 5580 (EDR, Logix capture)
ProSoft MVI56E-MNETC	*Client map limited to listed registers; no server function; configuration .ppf archived and protected.*	ProSoft MVI56E (EDR)
PowerFlex 755T	*CIP Security policy; communication-loss action Hold Last (OI-13); HIM access physically controlled; parameters in backup and change detection.*	PowerFlex 755T (EDR)
Danfoss FC 102	*SCF with named accounts and roles; 0-91 integrity check on; 0-92 hash recorded; SNMP (12-83) off unless used; Fire Mode (24-0*) disabled; 8-03 timeout 10 s and 8-04 Freeze output; audit log 43-6* forwarded where possible.*	Danfoss FC 102 (EDR)
Proline Promag 300/500	*User roles and web server restricted per vendor (OI-05); EtherNet/IP only to Z2 controller; DeviceCare backup.*	Promag P300/P500 (EDR)
PowerLogic PM8000	*Named accounts, defaults changed; Secure ION (7443) for ION Setup; HTTPS web; unused protocols off; BACnet read-only.*	PM8000 (EDR)
PowerLogic PM5560	*Firmware 4.x; administrator password changed; HTTPS only with port 80 blocked at the switch ACL; Modbus IP filtering on with two client addresses; display passcodes changed; banner.*	PM5560 (EDR)
QuickServer PS-QS-3x10-F	*Label password changed; HTTPS with owner certificate; users per group; cloud (Grid) registration off; firmware 7.0.0 or later; WAN rules restricted.*	QuickServer (NDR)
SIMATIC S7-1500 (skid)	*Access protection set; user management roles; secure PG/HMI communication; PUT/GET disabled; web server HTTPS only or off; OPC UA SignAndEncrypt, guest off; read-only Modbus server area (OI-15).*	S7-1500 (EDR)
Desigo PXC7	*BACnet/SC with owner certificates; local web access restricted to Z6; firmware current.*	Desigo PXC (EDR)
Stratix 5800	*AAA via TACACS+; SSH only; HTTP server off; SNMPv3; storm control; port security and unused ports shut in a parking VLAN; NTP authentication; syslog over TLS where supported; banner; exec-timeout; DLR and PTP per DD-05, DD-09.*	Stratix 5800 (NDR)
Catalyst 9300	*As Stratix 5800 for IOS-XE management; 802.1X on access ports in Z5 and Z6.*	Catalyst 9300 (NDR)
FortiGate (FortiOS 7.6)	*Deny-all policies per SDS 4; trusted hosts on admins; admin MFA; lockout threshold and duration; strong-crypto; pre-login banner; encrypted configuration backups; HA without bypass; island-mode policy sets; industrial signatures (OI-14).*	FortiGate (NDR)
Desigo CC V9 (≥ V9.0 QU1)	*Domain authentication; scope rights; encrypted client–server communication with owner certificates; OPC UA client SignAndEncrypt; activity log forwarded; cloud connectors off.*	Desigo CC (SAR)
Kepware	*Per HIS-04.*	Kepware (SAR)
Windows Server 2026 / Windows 11	*OT domain GPOs per SVC-01; BitLocker; Credential Guard; application control; SMBv1 off; RDP NLA; event forwarding.*	Windows Server 2026 / Windows 11 (HDR)
VMware ESXi 8	*Lockdown mode; SSH and shell off; Security.AccountLockFailures and AccountUnlockTime; Security.PasswordQualityControl; Syslog.global.logHost; Secure Boot with execInstalledOnly; TPM attestation; management only from Z7-M.*	VMware ESXi 8 (HDR)

10 Role and permission matrix

Function	Viewer	Plant Operator	Shift Supervisor	Plant Engineer	Controls Engineer	Security Admin	Vendor Service
View plant, alarms, trends	✓	✓	✓	✓	✓	✓	✓ (session)

Function	Viewer	Plant Operator	Shift Supervisor	Plant Engineer	Controls Engineer	Security Admin	Vendor Service
Acknowledge alarms		✓	✓	✓	✓		
Request setpoint / demand limit		✓	✓	✓	✓		
Enter Emergency mode			✓	✓	✓		
Manual unit control, maintenance lockout				✓	✓		
Shelve alarms			✓	✓			
Change controller logic or device configuration					✓		✓ (approved session)
Change firewall, switch, identity or PKI configuration						✓	

11 ISA/IEC 62443-3-3 system requirements allocation

Every system requirement at SL-T 3, with its requirement enhancements as stated in the URS, allocated to the design elements that meet it and the points where it is enforced.

Ref.	Requirement	Design elements	Enforcement points	URS	FRS
FR 1 — Identification and authentication control (IAC)					
SR 1.1	Human user identification and authentication	*SVC-01, DD-07, DD-11, BMS-02; HDS PH-05*	*All interactive interfaces; jump host MFA in Z8*	URS-S-101	FRS-SEC-101
SR 1.2	Software process and device identification and authentication	*DD-08, SVC-02, BMS-03, HIS-01, rule register (address pinning)*	*C1–C12; Z2 CIP Security*	URS-S-102	FRS-SEC-102
SR 1.3	Account management	*DD-07, SVC-01*	*Z7 directory; device account register*	URS-S-103	FRS-SEC-103
SR 1.4	Identifier management	*SVC-01*	*Z7 directory*	URS-S-104	FRS-SEC-104
SR 1.5	Authenticator management	*SVC-01, SVC-02, SDS 9 baselines; HDS HS-01*	*All components*	URS-S-105	FRS-SEC-105
SR 1.6	Wireless access management	*SDS 9 baselines; HDS PH-06*	*Z1–Z10*	URS-S-106	FRS-SEC-106
SR 1.7	Strength of password-based authentication	*SVC-01, SDS 9*	*Directory, FortiGate, switches, ESXi*	URS-S-107	FRS-SEC-107
SR 1.8	Public key infrastructure certificates	*DD-08, SVC-02*	*Z7 PKI; all TLS, OPC UA, BACnet/SC, CIP Security endpoints*	URS-S-108	FRS-SEC-108
SR 1.9	Strength of public key authentication	*DD-08, SVC-02*	*Z7 CRL*	URS-S-109	FRS-SEC-109
SR 1.10	Authenticator feedback	*SDS 9*	*All interfaces*	URS-S-110	FRS-SEC-110
SR 1.11	Unsuccessful login attempts	*SVC-01, SDS 9*	*Directory, FortiGate, ESXi, QuickServer, FC 102*	URS-S-111	FRS-SEC-111
SR 1.12	System use notification	*SVC-01, SDS 9*	*All supporting interfaces*	URS-S-112	FRS-SEC-112
SR 1.13	Access via untrusted networks	*DD-11, rules R-C6-01, R-C7-01*	*Z8 jump host; FortiGate 200F pair*	URS-S-113	FRS-SEC-113
FR 2 — Use control (UC)					
SR 2.1	Authorization enforcement	*SDS 10, BMS-02, SW-04, HIS-04*	*Every application and device*	URS-S-201	FRS-SEC-201
SR 2.2	Wireless use control	*SVC-07; HDS PH-06*	*Plant room and panels*	URS-S-202	FRS-SEC-202
SR 2.3	Use control for portable and mobile devices	*SVC-01; HDS PH-04, hardware features table*	*EWS, panels, controllers*	URS-S-203	FRS-SEC-203
SR 2.4	Mobile code	*SVC-01, SVC-05*	*Z5, Z6, Z7 hosts*	URS-S-204	FRS-SEC-204
SR 2.5	Session lock	*SVC-01*	*Workstations and consoles*	URS-S-205	FRS-SEC-205
SR 2.6	Remote session termination	*SVC-01, DD-11*	*Jump host, FortiGate, switches, ESXi*	URS-S-206	FRS-SEC-206
SR 2.7	Concurrent session control	*SVC-01, HIS-04*	*Controllers, FortiGate, applications*	URS-S-207	FRS-SEC-207
SR 2.8	Auditable events	*SVC-04, SW-07*	*All sources → Z7 collector*	URS-S-208	FRS-SEC-208
SR 2.9	Audit storage capacity	*SVC-04*	*Z7 collector*	URS-S-209	FRS-SEC-209
SR 2.10	Response to audit processing failures	*SVC-04, BMS-04*	*Z7 collector; operator alarms*	URS-S-210	FRS-SEC-210
SR 2.11	Timestamps	*DD-09, SVC-03; HDS NW-07*	*Z7 NTP; Z2 PTP*	URS-S-211	FRS-SEC-211
SR 2.12	Non-repudiation	*SW-05, SW-08, SVC-01*	*Controllers, BMS, firewall change records*	URS-S-212	FRS-SEC-212

Ref.	Requirement	Design elements	Enforcement points	URS	FRS
FR 3 — System integrity (SI)					
SR 3.1	Communication integrity	*DD-02, DD-03, BMS-03, HIS-01, protocol table*	*All conduits*	URS-S-301	FRS-SEC-301
SR 3.2	Malicious code protection	*SVC-05*	*Hosts; C6, C7*	URS-S-302	FRS-SEC-302
SR 3.3	Security functionality verification	*SVC-08*	*All components*	URS-S-303	FRS-SEC-303
SR 3.4	Software and information integrity	*SW-07, SW-08, SVC-08*	*Controllers, drives, hosts*	URS-S-304	FRS-SEC-304
SR 3.5	Input validation	*SW-04, SW-05*	*Plant controllers*	URS-S-305	FRS-SEC-305
SR 3.6	Deterministic output	*SW-06; HDS section 5*	*Final elements*	URS-S-306	FRS-SEC-306
SR 3.7	Error handling	*SDS 9*	*Web and OPC UA endpoints*	URS-S-307	FRS-SEC-307
SR 3.8	Session integrity	*SDS 9*	*Web and OPC UA endpoints*	URS-S-308	FRS-SEC-308
SR 3.9	Protection of audit information	*SVC-04*	*Z7 collector*	URS-S-309	FRS-SEC-309
FR 4 — Data confidentiality (DC)					
SR 4.1	Information confidentiality	*SVC-09, HIS-02*	*Backups, volumes, transport*	URS-S-401	FRS-SEC-401
SR 4.2	Information persistence	*SVC-09; HDS EN-05*	*Decommissioning*	URS-S-402	FRS-SEC-402
SR 4.3	Use of cryptography	*SVC-02, SVC-09, SDS 9*	*All TLS and OPC UA endpoints*	URS-S-403	FRS-SEC-403
FR 5 — Restricted data flow (RDF)					
SR 5.1	Network segmentation	*DD-04, DD-05, SDS 3*	*Physical plant network*	URS-S-501	FRS-SEC-501
SR 5.2	Zone boundary protection	*DD-04, rule register, SVC-10*	*FortiGate pairs*	URS-S-502	FRS-SEC-502
SR 5.3	General purpose person-to-person communication restrictions	*SVC-10*	*FortiGate application control*	URS-S-503	FRS-SEC-503
SR 5.4	Application partitioning	*DD-06, SDS 3*	*ESXi host pairs, VLANs*	URS-S-504	FRS-SEC-504
FR 6 — Timely response to events (TRE)					
SR 6.1	Audit log accessibility	*SVC-04*	*Z7 collector API*	URS-S-601	FRS-SEC-601
SR 6.2	Continuous monitoring	*SVC-07; HDS NW-05*	*SPAN or TAP sensors*	URS-S-602	FRS-SEC-602
FR 7 — Resource availability (RA)					
SR 7.1	Denial of service protection	*SVC-10, DD-05*	*Stratix storm control; FortiGate DoS*	URS-S-701	FRS-SEC-701
SR 7.2	Resource management	*SW-07, HIS-04*	*Controllers, hosts*	URS-S-702	FRS-SEC-702
SR 7.3	Control system backup	*SVC-06, SW-08*	*Z7 backup server*	URS-S-703	FRS-SEC-703
SR 7.4	Control system recovery and reconstitution	*SVC-06; HDS EN-04*	*Spares, gold images*	URS-S-704	FRS-SEC-704
SR 7.5	Emergency power	*SVC-06; HDS PW-01, PW-03*	*UPS, restart behaviour*	URS-S-705	FRS-SEC-705
SR 7.6	Network and security configuration settings	*SDS 9, SVC-08*	*All components*	URS-S-706	FRS-SEC-706
SR 7.7	Least functionality	*SW-09, SDS 9; HDS NW-03*	*All components*	URS-S-707	FRS-SEC-707
SR 7.8	Control system component inventory	*SVC-08; HDS EN-03*	*Z7 inventory*	URS-S-708	FRS-SEC-708

12 ISA/IEC 62443-4-2 SAR requirements for systems and software

Every software application in the system shall meet each requirement below at SL-C 3, including the requirement enhancements specified for SL 3, by capability or by the named compensation (URS-C-002). The host each application runs on is a separate HDR (HDS section 12).

12.1 Software inventory

Tag	Software	Zone	Checklist
SW-DCC	*Desigo CC V9 (≥ V9.0 QU1)*	Z5	Desigo CC (library)
SW-KEP	*Kepware with Kepware historian*	Z5	Kepware (library)
SW-S5K	*Studio 5000 Logix Designer, FactoryTalk Security, Policy Manager, Linx*	Z6	Studio 5000; FactoryTalk (library)
SW-FTV	*FactoryTalk View SE (local station and development)*	Z2, Z6	FactoryTalk (library)
SW-TIA	*TIA Portal (skid engineering)*	Z6 / vendor	New checklist needed
SW-TTU	*Trane Tracer TU*	Z6	New checklist needed
SW-MCT	*Danfoss MCT 10*	Z6	New checklist needed
SW-DVC	*Endress+Hauser DeviceCare / FieldCare*	Z6	New checklist needed
SW-ION	*Schneider ION Setup*	Z6	ION Setup (library)
SW-PCB	*ProSoft Configuration Builder*	Z6	New checklist needed
SW-ABT	*Siemens Desigo ABT (PXC engineering)*	Z6	New checklist needed
SW-OWN	*Owner services: log collector, monitoring console, backup, jump host and recording, endpoint protection, AD CS*	Z7, Z8	Per selected product

12.2 Requirements

Ref.	Requirement	Applies to	Design realization (worked example)	Software-specific gaps and compensation	FRS
Component requirements (all component types)					
CR 1.1	Human user identification and authentication	All SAR	*Desigo CC, Kepware, FactoryTalk and owner services authenticate named domain users; engineering tools run under named domain logons.*	*Vendor engineering tools without their own login (e.g., Tracer TU, MCT 10, DeviceCare, PCB) → host login plus device-side authentication.*	FRS-CMP-101
CR 1.2	Software process and device identification and authentication	All SAR	*OPC UA application certificates (Desigo CC, Kepware); BACnet/SC (Desigo CC); FactoryTalk Linx with CIP Security.*	*Tools using unauthenticated device protocols → Z6-only conduit (R-C3).*	FRS-CMP-102
CR 1.3	Account management	All SAR	*Accounts and groups in the OT domain; application roles mapped to groups.*	—	FRS-CMP-103
CR 1.4	Identifier management	All SAR	*Unique identifiers; service accounts per application.*	—	FRS-CMP-104
CR 1.5	Authenticator management	All SAR	*Service account passwords vaulted; certificates from owner PKI.*	—	FRS-CMP-105
CR 1.7	Strength of password-based authentication	All SAR	*Domain policy applies to application logons.*	—	FRS-CMP-107
CR 1.8	Public key infrastructure certificates	All SAR	*Owner-PKI certificates for Desigo CC, Kepware, FactoryTalk and owner services.*	—	FRS-CMP-108

Ref.	Requirement	Applies to	Design realization (worked example)	Software-specific gaps and compensation	FRS
CR 1.9	Strength of public key-based authentication	All SAR	*Trust lists and CRL checking in OPC UA and TLS stacks.*	—	FRS-CMP-109
CR 1.10	Authenticator feedback	All SAR	*Masked entry.*	—	FRS-CMP-110
CR 1.11	Unsuccessful login attempts	All SAR	*Domain lockout; application lockout where native.*	—	FRS-CMP-111
CR 1.12	System use notification	All SAR	*Host banner; application banner where supported.*	—	FRS-CMP-112
CR 1.14	Strength of symmetric key-based authentication	All SAR	*No hard-coded symmetric keys; secrets in application vaults.*	—	FRS-CMP-114
CR 2.1	Authorization enforcement	All SAR	*Role mapping per SDS 10 (Desigo CC scope rights, FactoryTalk Security permissions, Kepware user manager, read-only Kepware OPC UA user).*	—	FRS-CMP-201
CR 2.2	Wireless use control	All SAR	*Not applicable — no wireless function in the applications.*	—	FRS-CMP-202
CR 2.5	Session lock	All SAR	*Client session lock and timeout.*	—	FRS-CMP-205
CR 2.6	Remote session termination	All SAR	*Remote session termination via jump host and application timeouts.*	—	FRS-CMP-206
CR 2.7	Concurrent session control	All SAR	*Concurrent session limits per user.*	—	FRS-CMP-207
CR 2.8	Auditable events	All SAR	*Desigo CC activity log, FactoryTalk audit, Kepware event log, owner services logs → collector.*	*Engineering tools with limited logging → host logs and MOC records.*	FRS-CMP-208
CR 2.9	Audit storage capacity	All SAR	*Application log retention set; collector capacity.*	—	FRS-CMP-209
CR 2.10	Response to audit processing failures	All SAR	*Application keeps running if logging fails; alarm raised.*	—	FRS-CMP-210
CR 2.11	Timestamps	All SAR	*Host time from OT NTP.*	—	FRS-CMP-211
CR 2.12	Non-repudiation	All SAR	*Named users on every change; MOC comments.*	—	FRS-CMP-212
CR 3.1	Communication integrity	All SAR	*OPC UA SignAndEncrypt, BACnet/SC, TLS client–server encryption.*	—	FRS-CMP-301
CR 3.3	Security functionality verification	All SAR	*Configuration export and comparison.*	—	FRS-CMP-303
CR 3.4	Software and information integrity	All SAR	*Signed installers; file-integrity monitoring of application folders.*	—	FRS-CMP-304
CR 3.5	Input validation	All SAR	*Applications validate operator input; controller validates again (FRS-PLC-008).*	—	FRS-CMP-305
CR 3.6	Deterministic output	All SAR	*Applications fail without affecting plant control.*	—	FRS-CMP-306
CR 3.7	Error handling	All SAR	*Generic error messages.*	—	FRS-CMP-307
CR 3.8	Session integrity	All SAR	*Unique session identifiers, invalidated at logout.*	—	FRS-CMP-308
CR 3.9	Protection of audit information	All SAR	*Logs forwarded; local log access restricted.*	—	FRS-CMP-309
CR 4.1	Information confidentiality	All SAR	*Project, configuration and historian data on encrypted storage.*	—	FRS-CMP-401
CR 4.2	Information persistence	All SAR	*Application data purged at decommissioning.*	—	FRS-CMP-402
CR 4.3	Use of cryptography	All SAR	*Current algorithms only; deprecated OPC UA policies disabled.*	—	FRS-CMP-403
CR 5.1	Network segmentation	All SAR	*Applications installed only in their zone (SDS 3).*	—	FRS-CMP-501
CR 6.1	Audit log accessibility	All SAR	*Logs exportable programmatically.*	—	FRS-CMP-601
CR 6.2	Continuous monitoring	All SAR	*Application events monitored via collector.*	—	FRS-CMP-602

Ref.	Requirement	Applies to	Design realization (worked example)	Software-specific gaps and compensation	FRS
CR 7.1	Denial of service protection	All SAR	*Client connection limits; controller OPC UA load managed.*	—	FRS-CMP-701
CR 7.2	Resource management	All SAR	*Resource use monitored.*	—	FRS-CMP-702
CR 7.3	Control system backup	All SAR	*Project and database backups (SVC-06).*	—	FRS-CMP-703
CR 7.4	Control system recovery and reconstitution	All SAR	*Reinstall and restore runbooks.*	—	FRS-CMP-704
CR 7.6	Network and security configuration settings	All SAR	*Hardening per vendor guides and checklists.*	—	FRS-CMP-706
CR 7.7	Least functionality	All SAR	*Unused modules, drivers and interfaces removed.*	—	FRS-CMP-707
CR 7.8	Control system component inventory	All SAR	*Versions and patch levels in the inventory; SBOM where supplied.*	—	FRS-CMP-708
SAR-specific requirements					
SAR 2.4	Mobile code	All SAR	*Desigo CC and Kepware clients only on managed hosts.*	—	FRS-DEV-S204
SAR 3.2	Protection from malicious code	All SAR	*Desigo CC, Kepware and Studio 5000 run with vendor-qualified protection on their hosts.*	—	FRS-DEV-S302

Applies to: SW-DCC, SW-KEP, SW-S5K, SW-FTV, SW-TIA, SW-TTU, SW-MCT, SW-DVC, SW-ION, SW-PCB, SW-ABT, SW-OWN.

Annex A FRS → SDS coverage

Generated from the FRS and SDS data (and the HDS where present). FRS rows designed in neither document: none.

FRS row	SDS elements	HDS elements
FRS-CHL-001	SW-01, SW-05	—
FRS-CHL-002	SW-01	CH-n-EN, CH-n-RUN, FS-CHW-n / FS-CW-n
FRS-CHL-003	SW-01	—
FRS-CHL-004	SW-01	—
FRS-CHL-005	SW-01	—
FRS-CHL-006	DD-01, SW-01, SW-03	—
FRS-CHL-007	SW-01, SW-05	—
FRS-CHL-008	DD-03, SW-01	—
FRS-HYD-001	SW-01	P-CHW-1…4
FRS-HYD-002	SW-01	HDS-5 Minimum-flow bypass valve
FRS-HYD-003	SW-01	—
FRS-HYD-004	SW-01	—
FRS-HYD-005	SW-01	P-CW-1…4
FRS-CT-001	SW-01	F-CT-1…4
FRS-CT-002	SW-01	—
FRS-CT-003	SW-01	F-CT-1…4, VIB-CT-n
FRS-CT-004	SW-01	HDS-5 Tower bypass valve
FRS-ECO-001	SW-01	—
FRS-ECO-002	SW-01	HDS-5 Economizer HX valves
FRS-ECO-003	SW-01	—
FRS-TES-001	SW-01	TES-DV, HDS-5 TES discharge valve
FRS-TES-002	SW-01	—
FRS-TES-003	SW-01	—
FRS-WT-001	SW-10	WT-1
FRS-WT-002	SW-10	WT-1
FRS-PRH-001	DD-13	PXC-1
FRS-MET-001	SW-11	M-CH-1…4, M-MCC-1…4
FRS-MET-002	SW-02, SW-11	FT-CHW-1, 2
FRS-MET-003	SW-11, BMS-05	—
FRS-MOD-001	SW-06, SDS-10	—
FRS-MOD-002	SW-06	—
FRS-MOD-003	SW-06	—
FRS-MOD-004	SW-06	—
FRS-DEG-001	DD-10, SW-06	—
FRS-DEG-002	DD-12, SW-06, HIS-03	—
FRS-DEG-003	DD-01, SW-03, SW-06	CP-A, CP-B, PLC-A, PLC-B, HDS-3, PW-04
FRS-DEG-004	DD-05, SW-06	NW-01, EN-01
FRS-DEG-005	SW-06	UPS-C, UPS-ALM, PSU-ALM, PW-01, PW-02, PW-03
FRS-DEG-006	SW-06	TES-DV, HDS-5 Chiller isolation valves, HDS-5 Minimum-flow bypass valve, HDS-5 Economizer HX valves, HDS-5 TES discharge valve, HDS-5 Tower bypass valve
FRS-SAF-001	DD-13, SW-12	RM-1, RM-ALM
FRS-SAF-002	SW-12	—

FRS row	SDS elements	HDS elements
FRS-SAF-003	SW-12	FA-SD
FRS-SAF-004	SW-12	HDS-10 PowerFlex 755T / FC 102
FRS-SAF-005	SW-12	—
FRS-SAF-006	SW-12	CH-n-EN, FS-CHW-n / FS-CW-n, ES-PLANT
FRS-LCM-001	SW-13	—
FRS-LCM-002	SW-13	EN-04
FRS-LCM-003	SW-08, SW-13	—
FRS-LCM-004	R-C3-01, R-C3-02, SW-13	EWS-1, 2
FRS-HMI-001	DD-10	OIS-1, HDS-3
FRS-HMI-002	BMS-01, BMS-02, BMS-06	—
FRS-HMI-003	BMS-04	—
FRS-HMI-004	BMS-04, SDS-10	—
FRS-HMI-005	SW-07, BMS-04	—
FRS-HMI-006	DD-12, BMS-05	—
FRS-HMI-007	DD-02, DD-12, HIS-01, HIS-02	HS-02
FRS-HMI-008	BMS-05	—
FRS-INT-001	DD-03, R-C1-01, R-C1-02	MB-A, MB-B, HDS-3
FRS-INT-002	DD-03, R-C9-01, BMS-03	—
FRS-INT-003	DD-01, DD-02, R-C2-01, SW-04, BMS-03	PLC-A, PLC-B
FRS-INT-004	DD-02, R-C2-02, HIS-01	—
FRS-INT-005	R-C10-01, BMS-03, HIS-01	—
FRS-INT-006	R-C5-01	WT-TRIP
FRS-INT-007	R-C10-02, BMS-03	GW-1
FRS-INT-008	DD-13, R-C11-01, BMS-03	PXC-1
FRS-INT-009	R-C4-01, R-C4-02, BMS-03, HIS-01	—
FRS-INT-010	R-C3-01	EN-A1, EN-B1, P-CHW-1…4, HDS-3
FRS-INT-011	DD-11, R-C12-01, HIS-03	—
FRS-INT-012	—	WT-TRIP, HDS-4, PW-04
FRS-NET-001	DD-04, SDS-3, SVC-10	SW-C1, C2, NW-02, EN-03
FRS-NET-002	DD-04, R-DEF, SVC-10	FW-P1, P2, NW-04
FRS-NET-003	SVC-10	—
FRS-NET-004	DD-04, SVC-10	—
FRS-NET-005	DD-04, SVC-10	FW-P1, P2, FW-D1, D2, NW-04
FRS-NET-006	SVC-10	—
FRS-NET-007	DD-05, SDS-3, SVC-10	EN-A1, EN-B1, SW-R1…R6, NW-01, EN-02
FRS-NET-008	DD-06, R-C8-04, SVC-10	NW-06, HS-03
FRS-SEC-101	DD-07, BMS-02, SVC-01, SDS-11 SR 1.1	PH-05
FRS-SEC-102	SVC-01, SDS-11 SR 1.2	—
FRS-SEC-103	DD-07, R-C8-03, SVC-01, SDS-11 SR 1.3	—
FRS-SEC-104	DD-07, SVC-01, SDS-11 SR 1.4	—
FRS-SEC-105	SVC-01, SDS-11 SR 1.5	HS-01
FRS-SEC-106	SVC-01, SDS-11 SR 1.6	CH-1…4, PH-06
FRS-SEC-107	DD-07, SVC-01, SDS-11 SR 1.7	—
FRS-SEC-108	DD-08, SVC-01, SVC-02, SDS-11 SR 1.8	—
FRS-SEC-109	DD-08, SVC-01, SVC-02, SDS-11 SR 1.9	—

FRS row	SDS elements	HDS elements
FRS-SEC-110	SVC-01, SDS-11 SR 1.10	—
FRS-SEC-111	SVC-01, SDS-11 SR 1.11	—
FRS-SEC-112	SVC-01, SDS-11 SR 1.12	—
FRS-SEC-113	DD-11, R-C6-01, R-C7-01, SVC-01, SDS-11 SR 1.13	FW-D1, D2, JH-1
FRS-SEC-201	BMS-02, HIS-04, SVC-01, SDS-10, SDS-11 SR 2.1	—
FRS-SEC-202	SVC-07, SDS-11 SR 2.2	PH-06
FRS-SEC-203	SVC-01, SDS-11 SR 2.3	PH-01, PH-04, HDS-10 ControlLogix 5580
FRS-SEC-204	SVC-01, SDS-11 SR 2.4	—
FRS-SEC-205	SVC-01, SDS-11 SR 2.5	—
FRS-SEC-206	SVC-01, SDS-11 SR 2.6	—
FRS-SEC-207	SVC-01, SDS-11 SR 2.7	—
FRS-SEC-208	R-C8-02, SVC-04, SDS-11 SR 2.8	—
FRS-SEC-209	SVC-04, SDS-11 SR 2.9	—
FRS-SEC-210	SVC-04, SDS-11 SR 2.10	—
FRS-SEC-211	DD-09, R-C8-01, SVC-03, SDS-11 SR 2.11	SW-R1…R6, NTP-1, 2, NW-07
FRS-SEC-212	SW-05, SDS-11 SR 2.12	—
FRS-SEC-301	BMS-03, SDS-11 SR 3.1	—
FRS-SEC-302	R-C6-02, SVC-05, SDS-11 SR 3.2	—
FRS-SEC-303	SVC-08, SDS-11 SR 3.3	—
FRS-SEC-304	SW-07, SDS-11 SR 3.4	HDS-10 ControlLogix 5580
FRS-SEC-305	SW-04, SDS-11 SR 3.5	—
FRS-SEC-306	SW-06, SDS-11 SR 3.6	—
FRS-SEC-307	SVC-11, SDS-11 SR 3.7	—
FRS-SEC-308	SVC-11, SDS-11 SR 3.8	—
FRS-SEC-309	SVC-04, SDS-11 SR 3.9	—
FRS-SEC-401	HIS-02, SVC-09, SDS-11 SR 4.1	HS-02
FRS-SEC-402	SVC-09, SDS-11 SR 4.2	EN-05
FRS-SEC-403	SVC-02, SVC-09, SDS-11 SR 4.3	—
FRS-SEC-501	SDS-3, SVC-10, SDS-11 SR 5.1	—
FRS-SEC-502	DD-04, SVC-10, SDS-11 SR 5.2	—
FRS-SEC-503	SVC-10, SDS-11 SR 5.3	—
FRS-SEC-504	DD-06, BMS-01, SVC-10, SDS-11 SR 5.4	ESX-5A, 5B, ESX-7A, 7B
FRS-SEC-601	SVC-04, SDS-11 SR 6.1	—
FRS-SEC-602	SVC-07, SDS-11 SR 6.2	MON-1…n, NW-05
FRS-SEC-701	SVC-10, SDS-11 SR 7.1	—
FRS-SEC-702	SVC-10, SDS-11 SR 7.2	—
FRS-SEC-703	R-C8-05, SW-08, SVC-06, SDS-11 SR 7.3	—
FRS-SEC-704	SVC-06, SDS-11 SR 7.4	EN-04
FRS-SEC-705	SVC-06, SDS-11 SR 7.5	UPS-C, PW-01, PW-03
FRS-SEC-706	SVC-08, SDS-9, SDS-11 SR 7.6	—
FRS-SEC-707	SW-09, HIS-04, SDS-11 SR 7.7	NW-03
FRS-SEC-708	SVC-08, SDS-11 SR 7.8	EN-03
FRS-CMP-001	SVC-11, SDS-9	—
FRS-CMP-002	SVC-11, SDS-9	—
FRS-CMP-003	SVC-11, SDS-9	—

FRS row	SDS elements	HDS elements
FRS-CMP-004	DD-01, SVC-11, SDS-9	—
FRS-CMP-005	SVC-11, SDS-9	—
FRS-CMP-101	SVC-02, SDS-9, SDS-12 CR 1.1	PH-05, HDS-12 EDR CR 1.1, HDS-12 HDR CR 1.1, HDS-12 NDR CR 1.1
FRS-CMP-102	SVC-02, SDS-9, SDS-12 CR 1.2	HDS-12 EDR CR 1.2, HDS-12 HDR CR 1.2, HDS-12 NDR CR 1.2
FRS-CMP-103	SVC-02, SDS-9, SDS-12 CR 1.3	HDS-12 EDR CR 1.3, HDS-12 HDR CR 1.3, HDS-12 NDR CR 1.3
FRS-CMP-104	SVC-02, SDS-9, SDS-12 CR 1.4	HDS-12 EDR CR 1.4, HDS-12 HDR CR 1.4, HDS-12 NDR CR 1.4
FRS-CMP-105	SVC-02, SDS-9, SDS-12 CR 1.5	HDS-12 EDR CR 1.5, HDS-12 HDR CR 1.5, HDS-12 NDR CR 1.5
FRS-CMP-107	SVC-02, SDS-9, SDS-12 CR 1.7	HDS-12 EDR CR 1.7, HDS-12 HDR CR 1.7, HDS-12 NDR CR 1.7
FRS-CMP-108	DD-08, SVC-02, SDS-9, SDS-12 CR 1.8	HDS-12 EDR CR 1.8, HDS-12 HDR CR 1.8, HDS-12 NDR CR 1.8
FRS-CMP-109	DD-08, SVC-02, SDS-9, SDS-12 CR 1.9	HDS-12 EDR CR 1.9, HDS-12 HDR CR 1.9, HDS-12 NDR CR 1.9
FRS-CMP-110	SVC-02, SDS-9, SDS-12 CR 1.10	HDS-12 EDR CR 1.10, HDS-12 HDR CR 1.10, HDS-12 NDR CR 1.10
FRS-CMP-111	SVC-02, SDS-9, SDS-12 CR 1.11	HDS-12 EDR CR 1.11, HDS-12 HDR CR 1.11, HDS-12 NDR CR 1.11
FRS-CMP-112	SVC-02, SDS-9, SDS-12 CR 1.12	HDS-12 EDR CR 1.12, HDS-12 HDR CR 1.12, HDS-12 NDR CR 1.12
FRS-CMP-114	SVC-02, SDS-9, SDS-12 CR 1.14	HDS-12 EDR CR 1.14, HDS-12 HDR CR 1.14, HDS-12 NDR CR 1.14
FRS-CMP-201	SVC-04, SDS-9, SDS-12 CR 2.1	HDS-12 EDR CR 2.1, HDS-12 HDR CR 2.1, HDS-12 NDR CR 2.1
FRS-CMP-202	SVC-04, SDS-9, SDS-12 CR 2.2	HDS-12 EDR CR 2.2, HDS-12 HDR CR 2.2, HDS-12 NDR CR 2.2
FRS-CMP-205	SVC-04, SDS-9, SDS-12 CR 2.5	HDS-12 EDR CR 2.5, HDS-12 HDR CR 2.5, HDS-12 NDR CR 2.5
FRS-CMP-206	SVC-04, SDS-9, SDS-12 CR 2.6	HDS-12 EDR CR 2.6, HDS-12 HDR CR 2.6, HDS-12 NDR CR 2.6
FRS-CMP-207	SVC-04, SDS-9, SDS-12 CR 2.7	HDS-12 EDR CR 2.7, HDS-12 HDR CR 2.7, HDS-12 NDR CR 2.7
FRS-CMP-208	SVC-04, SDS-9, SDS-12 CR 2.8	HDS-12 EDR CR 2.8, HDS-12 HDR CR 2.8, HDS-12 NDR CR 2.8
FRS-CMP-209	SVC-04, SDS-9, SDS-12 CR 2.9	HDS-12 EDR CR 2.9, HDS-12 HDR CR 2.9, HDS-12 NDR CR 2.9
FRS-CMP-210	SVC-04, SDS-9, SDS-12 CR 2.10	HDS-12 EDR CR 2.10, HDS-12 HDR CR 2.10, HDS-12 NDR CR 2.10
FRS-CMP-211	DD-09, SVC-04, SDS-9, SDS-12 CR 2.11	HDS-12 EDR CR 2.11, HDS-12 HDR CR 2.11, HDS-12 NDR CR 2.11
FRS-CMP-212	SVC-04, SDS-9, SDS-12 CR 2.12	HDS-12 EDR CR 2.12, HDS-12 HDR CR 2.12, HDS-12 NDR CR 2.12
FRS-CMP-301	SVC-04, SDS-9, SDS-12 CR 3.1	HDS-12 EDR CR 3.1, HDS-12 HDR CR 3.1, HDS-12 NDR CR 3.1
FRS-CMP-303	SVC-04, SDS-9, SDS-12 CR 3.3	HDS-12 EDR CR 3.3, HDS-12 HDR CR 3.3, HDS-12 NDR CR 3.3
FRS-CMP-304	SVC-04, SDS-9, SDS-12 CR 3.4	HDS-12 EDR CR 3.4, HDS-12 HDR CR 3.4, HDS-12 NDR CR 3.4
FRS-CMP-305	SVC-04, SDS-9, SDS-12 CR 3.5	HDS-12 EDR CR 3.5, HDS-12 HDR CR 3.5, HDS-12 NDR CR 3.5
FRS-CMP-306	SVC-04, SDS-9, SDS-12 CR 3.6	HDS-12 EDR CR 3.6, HDS-12 HDR CR 3.6, HDS-12 NDR CR 3.6
FRS-CMP-307	SVC-04, SDS-9, SDS-12 CR 3.7	HDS-12 EDR CR 3.7, HDS-12 HDR CR 3.7, HDS-12 NDR CR 3.7

FRS row	SDS elements	HDS elements
FRS-CMP-308	SVC-04, SDS-9, SDS-12 CR 3.8	HDS-12 EDR CR 3.8, HDS-12 HDR CR 3.8, HDS-12 NDR CR 3.8
FRS-CMP-309	SVC-04, SDS-9, SDS-12 CR 3.9	HDS-12 EDR CR 3.9, HDS-12 HDR CR 3.9, HDS-12 NDR CR 3.9
FRS-CMP-401	SVC-09, SDS-9, SDS-12 CR 4.1	HDS-12 EDR CR 4.1, HDS-12 HDR CR 4.1, HDS-12 NDR CR 4.1
FRS-CMP-402	SVC-09, SDS-9, SDS-12 CR 4.2	EN-05, HDS-12 EDR CR 4.2, HDS-12 HDR CR 4.2, HDS-12 NDR CR 4.2
FRS-CMP-403	SVC-09, SDS-9, SDS-12 CR 4.3	HDS-12 EDR CR 4.3, HDS-12 HDR CR 4.3, HDS-12 NDR CR 4.3
FRS-CMP-501	SVC-10, SDS-9, SDS-12 CR 5.1	HDS-12 EDR CR 5.1, HDS-12 HDR CR 5.1, HDS-12 NDR CR 5.1
FRS-CMP-601	SVC-04, SDS-9, SDS-12 CR 6.1	HDS-12 EDR CR 6.1, HDS-12 HDR CR 6.1, HDS-12 NDR CR 6.1
FRS-CMP-602	SVC-04, SDS-9, SDS-12 CR 6.2	HDS-12 EDR CR 6.2, HDS-12 HDR CR 6.2, HDS-12 NDR CR 6.2
FRS-CMP-701	SVC-06, SDS-9, SDS-12 CR 7.1	HDS-12 EDR CR 7.1, HDS-12 HDR CR 7.1, HDS-12 NDR CR 7.1
FRS-CMP-702	SVC-06, SDS-9, SDS-12 CR 7.2	HDS-12 EDR CR 7.2, HDS-12 HDR CR 7.2, HDS-12 NDR CR 7.2
FRS-CMP-703	SVC-06, SDS-9, SDS-12 CR 7.3	HDS-12 EDR CR 7.3, HDS-12 HDR CR 7.3, HDS-12 NDR CR 7.3
FRS-CMP-704	SVC-06, SDS-9, SDS-12 CR 7.4	HDS-12 EDR CR 7.4, HDS-12 HDR CR 7.4, HDS-12 NDR CR 7.4
FRS-CMP-706	SVC-06, SDS-9, SDS-12 CR 7.6	HDS-12 EDR CR 7.6, HDS-12 HDR CR 7.6, HDS-12 NDR CR 7.6
FRS-CMP-707	SVC-06, SDS-9, SDS-12 CR 7.7	HDS-12 EDR CR 7.7, HDS-12 HDR CR 7.7, HDS-12 NDR CR 7.7
FRS-CMP-708	SVC-06, SDS-9, SDS-12 CR 7.8	HDS-12 EDR CR 7.8, HDS-12 HDR CR 7.8, HDS-12 NDR CR 7.8
FRS-DEV-E204	SVC-11, SDS-9	HDS-12 EDR 2.4
FRS-DEV-E213	SVC-11, SDS-9	CH-1…4, CP-n-DOOR, PH-02, PH-03, PH-04, HDS-10 ControlLogix 5580, HDS-10 Symbio 800, HDS-10 PowerFlex 755T / FC 102, HDS-12 EDR 2.13
FRS-DEV-E302	SVC-11, SDS-9	HDS-12 EDR 3.2
FRS-DEV-E310	SVC-08, SVC-11, SDS-9	HDS-12 EDR 3.10
FRS-DEV-E311	SVC-11, SDS-9	CP-n-DOOR, PH-01, PH-02, PH-03, HDS-12 EDR 3.11
FRS-DEV-E312	SVC-11, SDS-9	HDS-12 EDR 3.12
FRS-DEV-E313	SVC-11, SDS-9	HDS-12 EDR 3.13
FRS-DEV-E314	SVC-11, SDS-9	HDS-12 EDR 3.14
FRS-DEV-H204	SVC-11, SDS-9	HDS-12 HDR 2.4
FRS-DEV-H213	SVC-11, SDS-9	NW-06, HS-01, HS-03, HDS-12 HDR 2.13
FRS-DEV-H302	SVC-05, SVC-11, SDS-9	HDS-12 HDR 3.2
FRS-DEV-H310	SVC-08, SVC-11, SDS-9	HDS-12 HDR 3.10
FRS-DEV-H311	SVC-11, SDS-9	HS-04, HDS-12 HDR 3.11
FRS-DEV-H312	SVC-11, SDS-9	HS-01, HDS-10 Servers and workstations, HDS-12 HDR 3.12
FRS-DEV-H313	SVC-11, SDS-9	HDS-12 HDR 3.13
FRS-DEV-H314	SVC-11, SDS-9	ESX-5A, 5B, HS-01, HDS-10 Servers and workstations, HDS-10 Panel PC (OIS-1), HDS-12 HDR 3.14
FRS-DEV-N106	SVC-10, SDS-9	PH-06, HDS-12 NDR 1.6
FRS-DEV-N113	SVC-10, SDS-9	HDS-12 NDR 1.13
FRS-DEV-N204	SVC-10, SDS-9	HDS-12 NDR 2.4

FRS row	SDS elements	HDS elements
FRS-DEV-N213	SVC-10, SDS-9	NW-03, PH-04, HDS-10 Stratix 5800, HDS-10 FortiGate, HDS-12 NDR 2.13
FRS-DEV-N302	SVC-10, SDS-9	HDS-12 NDR 3.2
FRS-DEV-N310	SVC-08, SVC-10, SDS-9	HDS-12 NDR 3.10
FRS-DEV-N311	SVC-10, SDS-9	HS-04, PH-02, HDS-12 NDR 3.11
FRS-DEV-N312	SVC-10, SDS-9	HDS-12 NDR 3.12
FRS-DEV-N313	SVC-10, SDS-9	HDS-12 NDR 3.13
FRS-DEV-N314	SVC-10, SDS-9	HDS-12 NDR 3.14
FRS-DEV-N502	SVC-10, SDS-9	HDS-12 NDR 5.2
FRS-DEV-N503	SVC-10, SDS-9	HDS-12 NDR 5.3
FRS-DEV-S204	HIS-04, SDS-9, SDS-12 SAR 2.4	—
FRS-DEV-S302	HIS-04, SDS-9, SDS-12 SAR 3.2	—
FRS-PLC-001	SW-02	—
FRS-PLC-002	SW-07	HDS-10 ControlLogix 5580
FRS-PLC-003	SW-09	—
FRS-PLC-004	SW-03	—
FRS-PLC-005	SW-07, SW-08	—
FRS-PLC-006	SW-02	—
FRS-PLC-007	SW-02	—
FRS-PLC-008	SW-04, SW-05	—
FRS-PLC-009	SW-09	—
FRS-PLC-010	SW-04	—
FRS-PLC-011	SW-02	—
FRS-PLC-012	SW-02	—
FRS-PLC-013	SW-09	—
FRS-PLC-014	SW-04, SW-09	—
FRS-PLC-015	SW-01, SW-06	—
FRS-PLC-016	SW-01, SW-07	—
FRS-PLC-017	SW-07	—
FRS-PLC-018	SW-01, SW-07	—
FRS-PLC-019	SW-07	—
FRS-PLC-020	SW-02	—
FRS-PLC-021	SW-10	—

Annex B Open items to confirm

Carried from URS and FRS Annex D (OI-01 to OI-18) plus:

ID	Party	Item
OI-19	Siemens	Desigo CC V9 supported standby or redundancy configuration for the Z5 server.
OI-20	Vendors	Network ports used by Tracer TU, MCT 10, DeviceCare, ION Setup and TIA Portal secure communication, for rule R-C3-02.